Lec 5, 9/16/26

Quotient Space

- $\text{Vect} = \text{Vect}_{\mathbb{F}}^{f.d.}$.
- $U < V$ subspace.

Q. Is there a space W, and an onto map

$$V \xrightarrow{\varepsilon} W \to 0$$

s.t. $0 \to U \xrightarrow{i} V \xrightarrow{\varepsilon} W \to 0$ s.e.s.?

so W is a space with onto map $V \xrightarrow{\varepsilon} W$, $\ker(\varepsilon) = U$.

Def. Such a space W is called the <u>quotient space</u> of V modulo U, denoted $W = V/U$. $\dim < \infty$ in case U, V f.d.

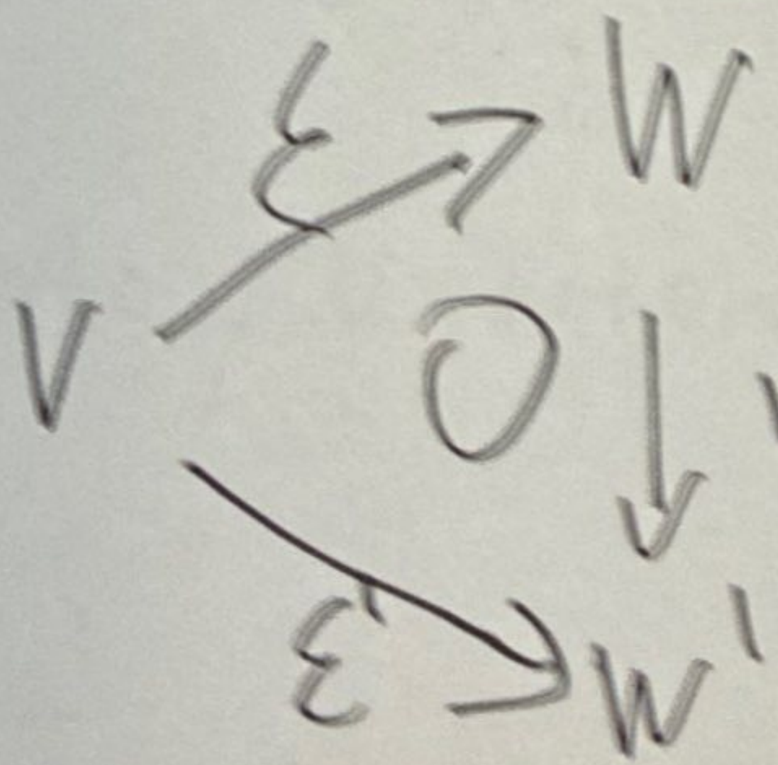
<u>Note:</u> $\dim(V/U) = \dim(V) - \dim(U)$.

Thm. We have:

(i) If $V \xrightarrow{\varepsilon} W$, $V \xrightarrow{\varepsilon'} W'$,

are two such quotients,

the $\exists!$



s.t.

$$v \circ \varepsilon = \varepsilon'.$$

Moreover such v is an isomorphism.

(ii) such $V \xrightarrow{\varepsilon} W$, exists.

(i) Pf.

Uniqueness: U must satisfy

$$U(\varepsilon(v)) = \varepsilon'(v),$$ and b.c.

- ε is onto, U determined.

Existence:

Define

$$U(w) = \varepsilon'(v), \text{ if } \varepsilon(v) = w.$$

Note: U well defind.

For every $v \in V$, denote

$$(v, u) \longmapsto v+u.$$

$$\Lambda \longleftarrow \cap \times \Lambda$$

$$\cap \subseteq \Lambda$$

(!) Consideration:

So $\varepsilon'(v) = \varepsilon'(\tilde{v})$.

$$\Omega \ni \tilde{\Lambda} - \Lambda \Leftarrow \varepsilon(\tilde{\Lambda}) = m = \varepsilon(\Lambda) \quad \text{If}$$

Why?

by $v+U$ its orbit under this action, called also (right) co-set.

Rk: $v+U = \tilde{v}+U \iff v-\tilde{v} \in U$.

Denote

V/U = space of orbits, (cosets).

we have a natural map

$V \xrightarrow{\pi} V/U$ - quotient map -

Note that:

$0 + U = U$, is special/ co-set.

Want: $+_{V/U}$ & $\cdot_{V/U}$ s.t Q

is L.T.

Need: $\forall\ v, v' \in V,\ \alpha \in \mathbb{F}$,

$$\begin{cases} Q(v+v') = Q(v) +_{V/U} Q(v') \\ Q(\alpha \cdot v) = \alpha \cdot Q(v). \end{cases}$$

so we must define:

$$\begin{cases} (v+U) +_{V/U} (v'+U) = (v+v') + U, \\ \alpha \cdot_{V/U} (v+U) = (\alpha \cdot v) + U. \end{cases}$$

claim. $+_{V/U}$ & $\cdot_{V/U}$ are well defined, and with $0_{V/U} = U$,

$$(V/U, +_{V/U}, 0_{V/U}, \cdot_{V/U})$$

is a V.S. over $\mathbb{F}$ &

the map

$$V \xrightarrow{q} V/U, \text{ is L.T.,}$$

with $\ker(q) = U$.

So $W = V/U$ is quotient

space of V mod U.

More next time.